

Inventory Optimization — User Journey, Setup & Reports (v1)

Product: Modisoft Back Office → Inventory Optimization

Scope: Read-only insights that leverage counts + DF forecasts. No auto-writes or agentic actions in v1. Sunny chat can surface the same insights via read endpoints.

1) End-to-End User Journey (Back Office)

A. Entry & Auth

1. User navigates to **Back Office** → logs in (SAML/Email+OTP).
2. RBAC applies **store scope** and **role** (Owner/Manager/Staff).
3. Home → **Inventory Optimization** appears under AI suite.

B. First-Run (if no prior use)

- Banner: short intro explaining “What you have / What you can sell,” horizons (6h/1d/3d), and data-health flags.
- CTA: **Run Setup Wizard** (5 steps, ~2 minutes) → see Section 2.

C. Daily Use Flow

1. **Choose context** (top bar): Business Type, Store(s), Department, Horizon (6h/1d/3d), Search.
2. Review **KPIs** and **AI Insights** (2–3 plain-English suggestions).
3. Scan **What you have** and **What you can sell** (toggle List/Bar/Pie).
4. Open **Quick Cycle Count** list for stale/negative items with Suggested On-Hand (SOH) and reason line.
5. Open **Auto-Replenishment** to see suggested order quantities (rule-based, case rounded).

6. Optional: click **Export / Print** or **Share to Sunny** (copies a link to the same filter state).
7. **Reports** → Month End / MoM / YoY (Section 4).

D. Sunny Chat Parity

- In Sunny: “What should I fix in beverages at Store 101 today?” → Sunny calls the same backend /invopt endpoints and returns the same insights as text.
(Read-only.)

2) Setup Wizard (Back Office)

Goal: finish in ~2 minutes, defaults are safe.

Step 1 — Stores & Business Type

- Multi-select stores.
- Business type per store (Convenience, Grocery, Liquor, Restaurant).
- Default target days by type (editable later): Conv 3, Grocery 4, Liquor 7, Restaurant 2.

Step 2 — Departments & Case Packs

- Select departments to include (default: all).
- Set **case pack source**: Item Master (default) or Vendor override.
- Validate that each SKU has a case pack; show a fix list if missing.

Step 3 — Forecast & Horizon

- Confirm you are connected to **Demand Forecasting** tables.
- Operational horizons available: **6h / 1d / 3d**.
- Historical presets (for charts/reports): **7d / 14d / 28d**.

Step 4 — Data-Health Policy

- Stale threshold: default **14 days** (editable).
- Negative OH handling: show in Cycle Count list and in Insights.
- Choose who sees data-health warnings (Owner/Manager/All staff).

Step 5 — Review & Save

- Summary card shows: store scope, depts, case pack policy, horizons, stale threshold.
- Finish → lands on Inventory Optimization with those defaults applied.

Wizard Telemetry: start/complete rate, average time, drop step.

3) Workflow Details & States

Filters: Business Type, Stores (multi), Dept, Horizon (6h/1d/3d), Search.

Views: List, Bar, Pie (both widgets).

Flags: stale (>14d since last count), suspect (negative OH).

Suggested OH:

- Primary: $SOH = \max(0, \text{LastPurchaseQty} - \text{DaysSincePurchase} \times \text{DailyPace})$; fallback to DailyPace (DF 1-day proxy).
- Show one-line **reason** (last purchase date/qty, pace, days).

Auto-Replenishment:

`OrderQty = ceil_to_case( max(0, targetDays × DailyPace - (OnHand + OnOrder)) )`

- Rule only. No lead-time modeling in v1.
- Button opens **Purchasing** with a prefilled cart payload (no auto-PO).

AI Insights (LLM):

- GPT-4.1-mini generates 2–3 items using a strict JSON schema from provided facts.
- Examples: “Shelf-out likely: Pepsi 20oz short by 8 today” / “Stale counts in Beverages (21d).”
- No free-form math, no writes.

Empty / Error states:

- No rows → “No items match filters.”
- Forecast missing → show counts only; insights omit shelf-out logic.
- DF down → banner + link to DF status.
- Permission error → ask admin to grant store scope.

4) Reports (EoM, MoM, YoY)

4.1 Report Types

- **End of Month (EoM)** — single month snapshot with comparisons to prior period.
- **Month-over-Month (MoM)** — change vs previous month.
- **Year-over-Year (YoY)** — change vs same month last year.

4.2 Metrics (by default)

Inventory Position

- Average On-Hand Units & \$
- Ending On-Hand Units & \$
- On-Order Units & \$
- Coverage ($\text{On-Hand} \div \text{Forecast}_h$)

Availability & Risk

- Items at Risk (count)
- Shelf-out Incidents (item-days where $F_h > \text{OnHand}$)
- Fill Rate ($\text{fulfilled} \div \text{forecast demand}$)
- Stock-outs (count of $\text{OnHand} \leq 0$ occurrences)

Data Health

- Stale Count Rate (% of items > 14d)
- Negative OH Incidents
- Corrected Items (from Cycle Count) — count

Movement / Velocity

- Daily Pace (avg) by Dept (units/day)
- Slow-Mover Dollars (estimate)

Ordering

- Suggested Order Units (sum, by dept)
- Case Efficiency (suggested ÷ case pack multiple)

All metrics available by **Store**, **Department**, **Vendor**, and **ABC class**.

4.3 Report Layout (print/PDF)

Header

- Company, Store(s), Dept filter, Period, Generated on (timestamp).
- KPIs row: Items at Risk, Fill Rate, Slow-Mover \$, Stale Rate, Stock-outs.

Section A — What You Have (Month View)

- Bar chart: OH vs Forecast (top 20 items).
- Table: UPC, Item, On-Hand avg, On-Order avg, Coverage, Quality.

Section B — What You Can Sell (Month View)

- Bar chart: Sellable vs Forecast.
- Table: UPC, Item, Forecast_h, Sellable, Shortfall, Shelf-out flag.

Section C — Data Health

- Pie: Good vs Stale vs Suspect.
- Table: items with last count date, negative OH count, Suggested OH.

Section D — Ordering Summary

- Suggested Order Units by Dept (stacked bar).
- Top 20 lines with case packs and order suggestions.

Footer

- Notes on methods; link to DF module if a user wants to adjust forecasts.

4.4 Report Export

- **Export formats:** PDF, CSV (detailed tables), XLSX.
- **Schedule:** optional email scheduling (monthly) — phase 2.

5) Report API Contracts

GET /invopt/report/eom

Params: store_ids[], dept?, month=YYYY-MM, currency=USD

Returns:

```
{
  "header": {"stores":["c101"], "dept":"Beverages", "period":"2025-10", "generated_at":"ISO"},
  "kpis": {"items_at_risk":12, "fill_rate":0.93, "slow_mover_usd":24800, "stale_rate":0.18, "stockouts":7},
  "have": [{"upc":"049000050103", "name":"Coca-Cola 20oz", "oh_avg":68, "on_order_avg":20, "coverage":1.2, "quality":"good"}],
  "sell": [{"upc":"049000050103", "name":"Coca-Cola 20oz", "forecast":160, "sellable":160, "shortfall":0, "shelf_out":false}],
  "health": {"good":120, "stale":24, "suspect":6, "items":[{"upc":"099999999999", "name":"Hot Dog Bun 8ct", "last_count":"2025-09-01", "oh":-3, "suggested_oh":6}]},
  "ordering": {"by_dept":[{"dept":"Beverages", "units":3200}], "top_lines":[{"upc":"012000809177", "name":"Pepsi 20oz", "case":24, "suggested":72}]}
}
```

GET /invopt/report/compare

Params: store_ids[], dept?, period_a=YYYY-MM, period_b=YYYY-MM, mode=MoM|YoY

Returns delta fields for KPIs and each table section (positive/negative arrows).

6) Analytics & Events

- `invopt_view {stores, dept, horizon, user_id}`
- `invopt_insight_rendered {insight_id, items, horizon}`
- `invopt_cyclecount_view {rows}`
- `invopt_autopo_view {rows, target_days}`
- `invopt_report_export {type: PDF|CSV|XLSX, kbis}`
- `invopt_share_to_sunny {stores, dept, horizon}`

7) QA Acceptance (for this doc)

- Wizard completes with defaults; Case Pack warnings appear if any SKU missing pack size.
- Reports generate for EoM, MoM, YoY with correct header context and matching dashboard numbers for the same period/filters.
- Sunny chat returns the same top 3 insights for identical filters (text, not charts).
- Exported PDF matches layout spec; CSV/XLSX include the raw table rows.

8) Open Decisions

1. Confirm **stale threshold** default (14d) and whether it's global or per-dept.
2. Confirm **case pack** precedence: item master vs vendor override.
3. Do we want **email scheduling** for EoM reports in phase 1 or 2?
4. Any branded PDF header/footer requirements from Marketing?